DATA WRANGLING GROUP PROJECT

Breathe India

What really drives air pollution in Indian cities?

26
cities

5½ years
of daily data, 2015 to 2020

4
sources joined

3
machine-learning models

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THE ANSWER, UP FRONT

A third of the lockdown's “clean air” was actually the weather

−41%

PM2.5 change in spring 2020 vs 2019: the headline everyone quoted

−28%

the real lockdown effect once our model removes the weather

75%

of next-day air-quality categories predicted exactly (baseline 72%)

Also: cleaning changed the story. Officially, Ahmedabad is India's most polluted city. It isn't: a sensor column was copied.

DATA AND PIPELINE

Four sources, one star schema, three storage formats

CPCB air quality
Kaggle download · 29,531 city-days, 108,035 station-days

Open-Meteo weather API
52,234 days of ERA5 weather · rate-limited, cached

Open-Meteo geocoding API
26 cities located · names checked by state

Event calendar
Lockdown phases and Diwali dates

Clean fact table
28,260 city-days × 62 columns

23 logged cleaning decisions

Validated joins, lags, rolling means

SQLite
keys + read-back checks

Parquet
partitioned by year

Power BI
5-page dashboard

Models A, B, C
scikit-learn

Everything reruns end to end in about 8 minutes from one notebook: github.com/mahak2604/breathe-india

CLEANING

The raw data was wrong in ways a quick look would miss

1 A copied column

Ahmedabad's CO equals its NO on 94% of days, although the units differ 1000×. It drove 95% of the city's “Severe” days.

2 Impossible pairs

PM2.5 above PM10 on 256 days. We assumed PM10 was wrong; the data showed PM2.5 was the faulty one 94% of the time.

3 Broken sensors

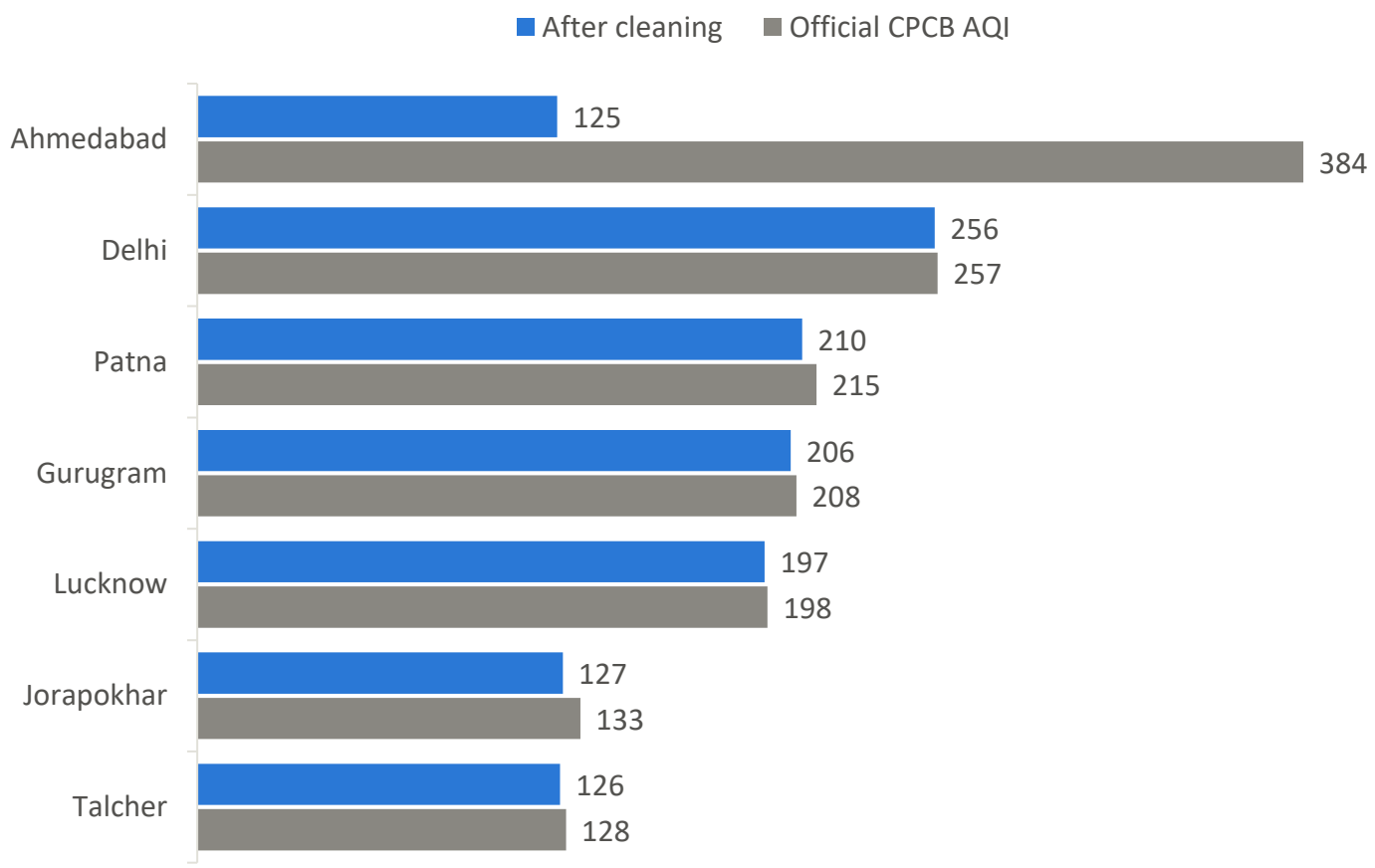
A sensor stuck at 11.00 for 8 days, and 917 µg/m³ in Shillong, the cleanest city. 197 readings removed, with Diwali spikes kept.

4 Gaps and limits

AQI up to 2,049 on a 0–500 scale, exact zeros, 61% of one column empty. Short gaps interpolated, the rest left honest.

INSIGHT 1

Cleaning moved Ahmedabad from India's most polluted city to 7th



Median daily AQI, 2015–2020, top 7 cities on official numbers

384 → 125

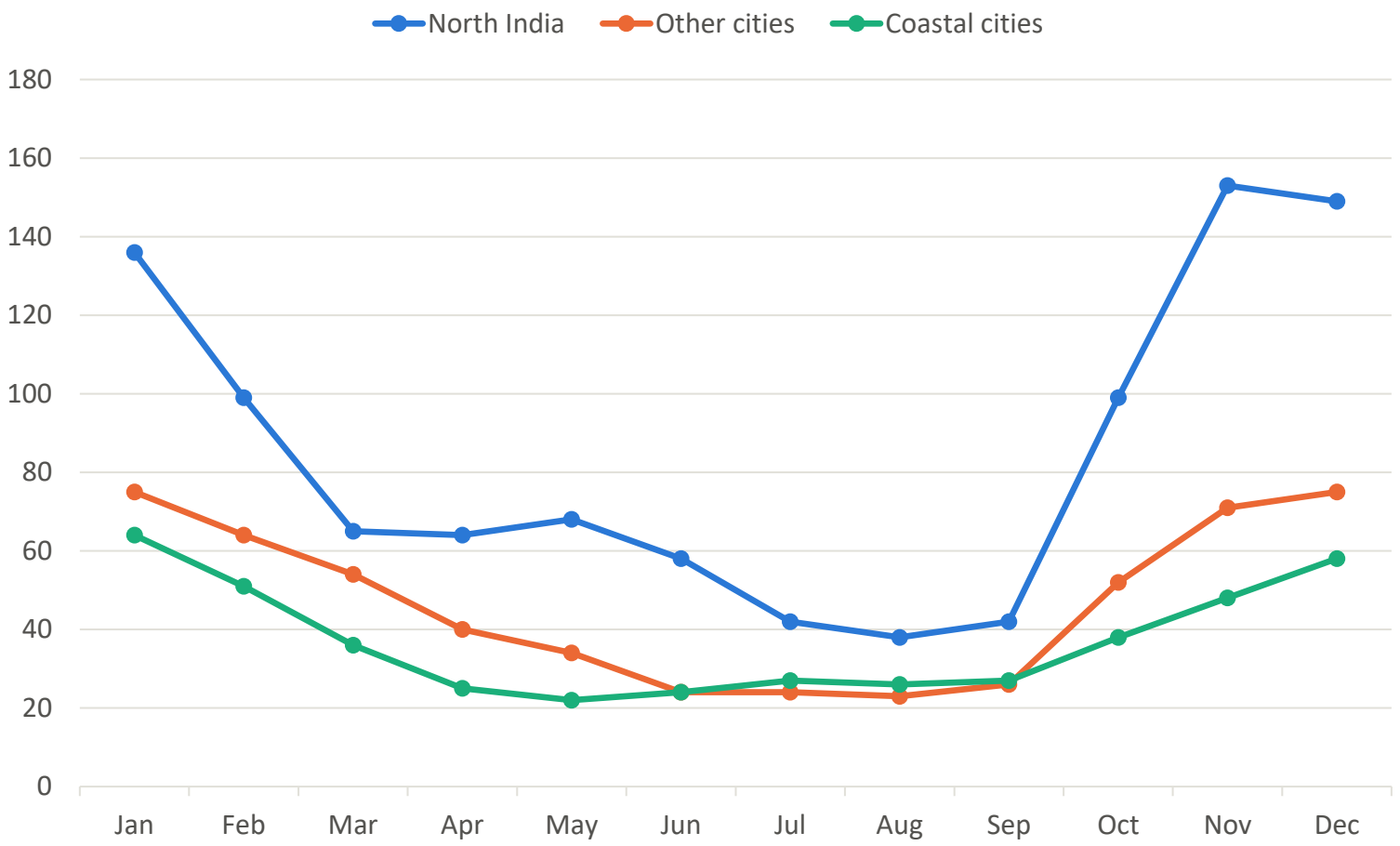
Ahmedabad's median AQI, official vs cleaned

Every other city barely moves.

Without this one fix, any ranking, map or model would have named the wrong city.

INSIGHTS 2 AND 3

North India's winter air carries 3.3× the PM2.5 of its monsoon air



Median PM2.5 (µg/m³) by month, 2015–2020 · India's 24-hour limit is 60

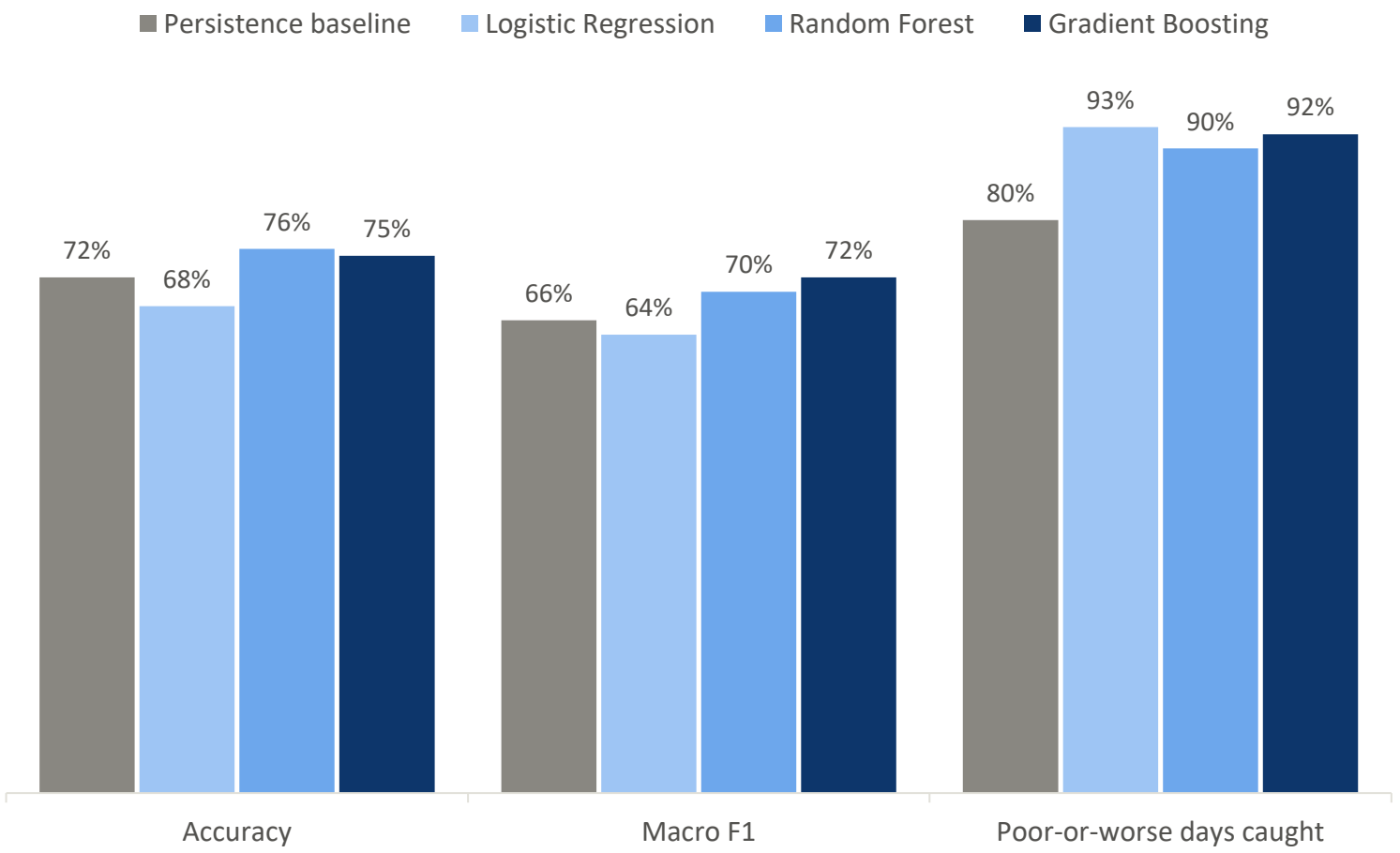
148 vs 44

µg/m³: North India in Nov–Jan vs Jun–Sep. Coastal cities swing only 2.2×.

23%, not 65%

how much cleaner a heavy-rain day really is, once we compare the same city in the same month

Model A warns a day ahead and catches 92% of Poor-or-worse days

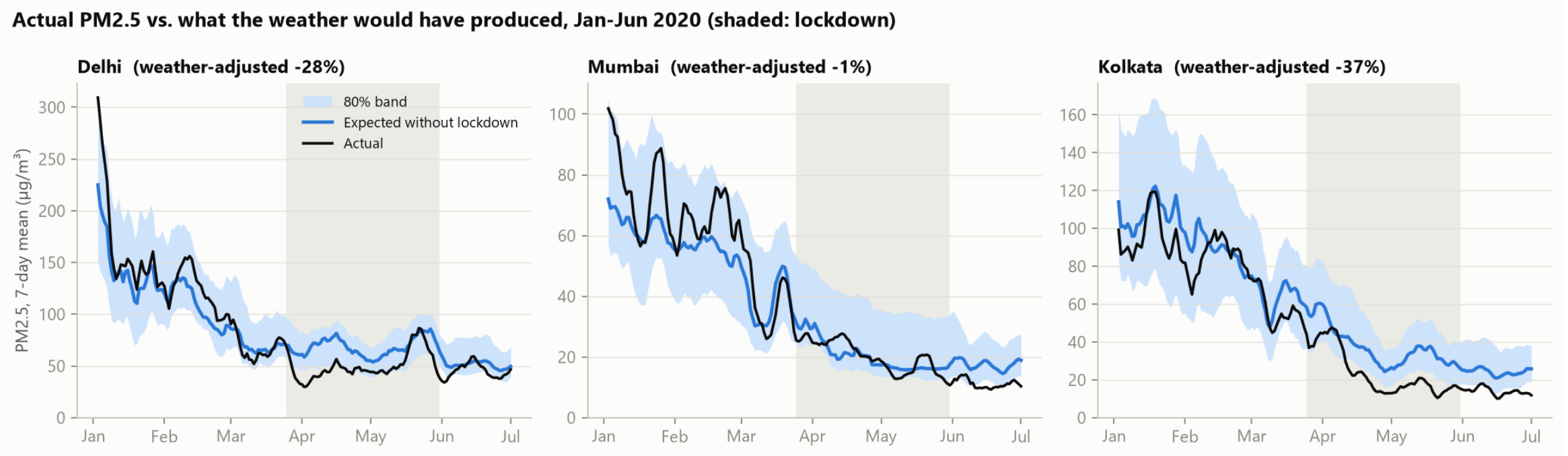


Gradient Boosting beats "same as today" on accuracy (75% vs 72%) and is right 35% of the time when the category changes, where the baseline is always wrong.

Honest caveat: in the lockdown its macro F1 fell to 0.59, below the baseline's 0.65. The rules it learned stopped applying.

Unseen test period Jan 2019 – Mar 2020 · time-ordered split, time-series cross-validation

Weather removed, the lockdown cut PM2.5 by 28%, not 41%



–41% → –28%

PM2.5: naive vs weather-adjusted, median of 17 cities

–47% → –46%

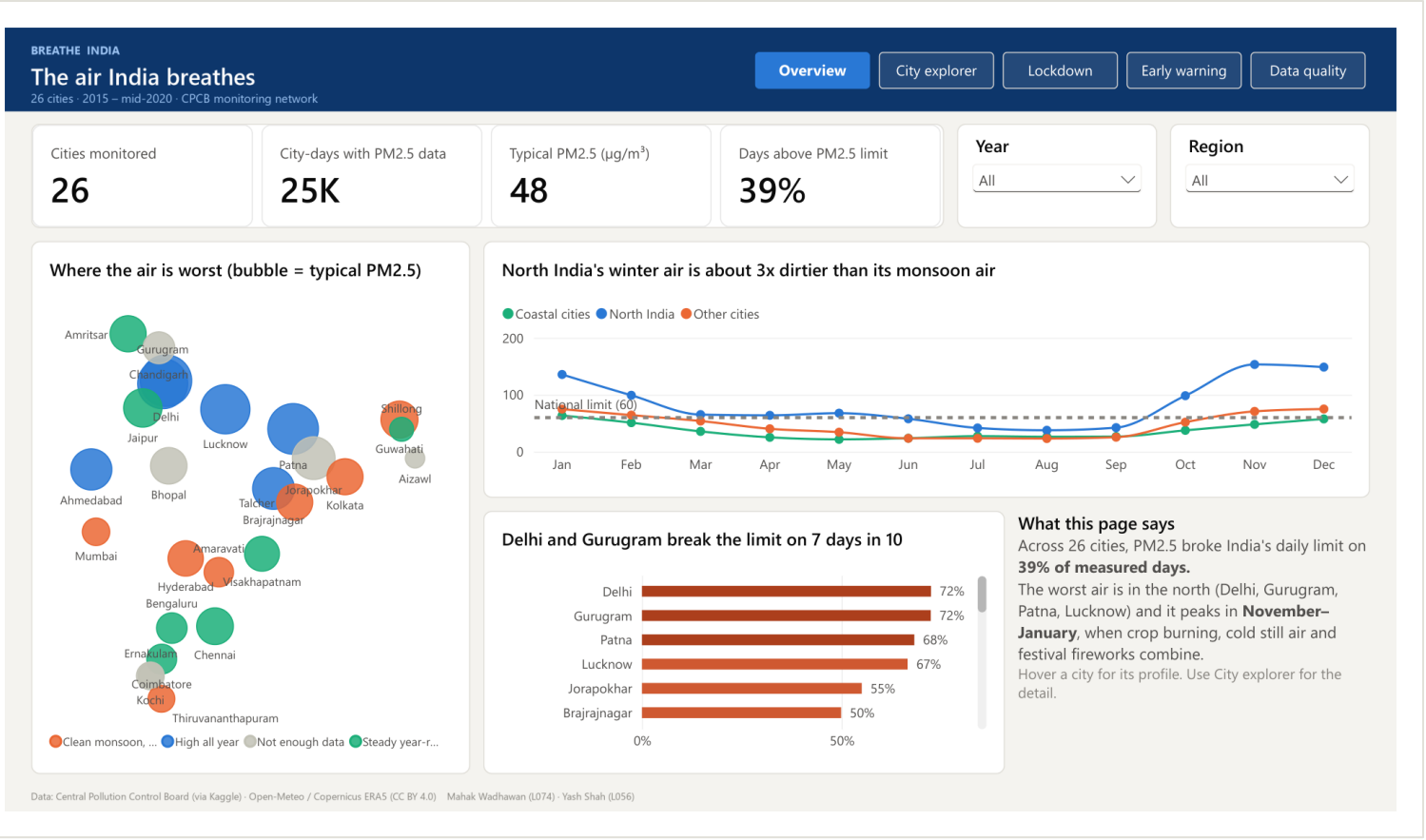
NO_2 (mostly traffic): a real drop either way

R^2 0.55

placebo test on spring 2019, when nothing happened

POWER BI

The dashboard turns this into five questions a non-expert can explore



Live demo: Breathe_India.pbix · hover a city, drill into Delhi, switch Test → Lockdown

1. **Overview** — where and when the air is worst
2. **City explorer** — one city's year, stations, AQI bands
3. **Lockdown** — actual vs expected, pick any city
4. **Early warning** — model vs baseline, switch periods
5. **Data quality** — every cleaning decision and why

What we can say, and what we can't

Limitations

Monitors cover parts of 26 cities, not everyone's exposure

Our AQI corrections are recalculations, not CPCB's own

Model A uses actual weather as a forecast stand-in

Model B assumes 2015–19 “normal” would have held in 2020

Data ends 1 July 2020

Takeaways

Season sets the baseline: winter is the problem in the north

People still matter: −28% PM2.5 and −46% NO_2 without traffic

Wrangling changed the answer: check the data before trusting a ranking

Contribution: Mahak led extraction, cleaning and storage; Yash led transformation, the models and the dashboard; insights, report and deck together.

Thank you. Questions?